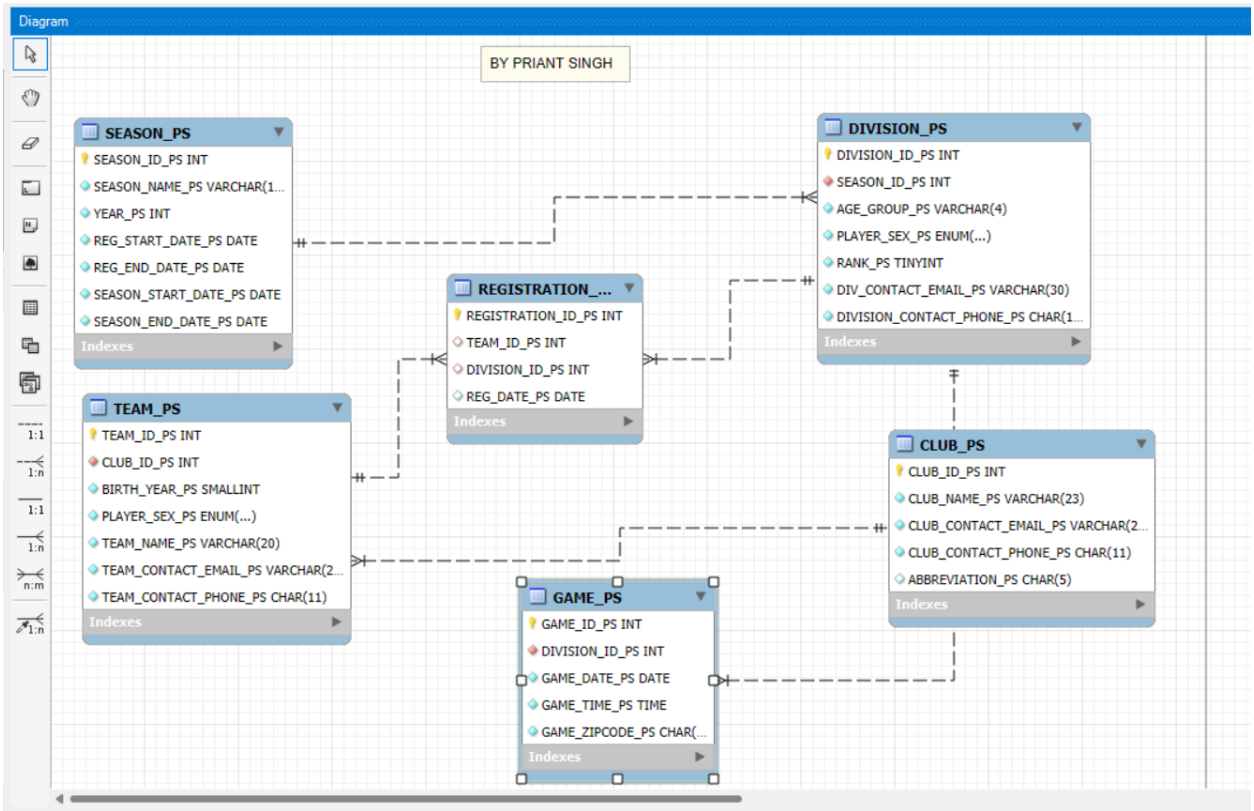
S5 SQL Implementation Assignment

PRIANT SINGH

TASK 1 – ERD MODEL



TASK 2 – CREATE DATABASE

SQL File 3* | club_ps | division_ps | registration_ps | season_ps | team_ps | SQL File 8*

Limit to 1000 rows

```sql
-- Task 2 - Create database by Priant Singh
DROP DATABASE IF EXISTS soccer_league_ps;
CREATE DATABASE soccer_league_ps;
USE soccer_league_ps;
SHOW DATABASES;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Database
hr
information_schema
mysql
performance_schema
soccer_league_ps
sys

Result 7 | Result 8 | Result 9 | Result 10

Output

Action Output

#	Time	Action	Message
1	11:00:28	DROP DATABASE IF EXISTS soccer_league_PS	5 row(s) affected
2	11:00:28	CREATE DATABASE soccer_league_PS	1 row(s) affected
3	11:00:28	USE soccer_league_PS	0 row(s) affected
4	11:00:28	SHOW DATABASES	6 row(s) returned

TASK 3 – CREATE TABLES

```sql
-- Task 3 - Create tables by Priant Singh
SHOW TABLES;
-- S3 tables
CREATE TABLE CLUB_PS (
  CLUB_ID_PS                INT           NOT NULL,
  CLUB_NAME_PS              VARCHAR(50)   NOT NULL,
  CLUB_CONTACT_EMAIL_PS     VARCHAR(50),
  CLUB_CONTACT_PHONE_PS     CHAR(11),
  ABBREVIATION_PS           CHAR(5),
  PRIMARY KEY (CLUB_ID_PS)
);

CREATE TABLE SEASON_PS (
  SEASON_ID_PS              INT           NOT NULL,
  SEASON_NAME_PS            VARCHAR(10)   NOT NULL,
  YEAR_PS                   INT           NOT NULL,
  REG_START_DATE_PS         DATE,
  REG_END_DATE_PS           DATE,
  SEASON_START_DATE_PS      DATE,
  SEASON_END_DATE_PS        DATE,
```

```sql
  PRIMARY KEY (SEASON_ID_PS)
);

CREATE TABLE DIVISION_PS (
  DIVISION_ID_PS            INT           NOT NULL,
  SEASON_ID_PS              INT           NOT NULL,
  AGE_GROUP_PS              VARCHAR(4),
  PLAYER_SEX_PS             ENUM('M','F'),
  RANK_PS                   TINYINT,
  DIV_CONTACT_EMAIL_PS      VARCHAR(50),
  DIVISION_CONTACT_PHONE_PS CHAR(11),
  PRIMARY KEY (DIVISION_ID_PS),
  CONSTRAINT FK_DIVISION_PS_SEASON_PS
    FOREIGN KEY (SEASON_ID_PS)
    REFERENCES SEASON_PS(SEASON_ID_PS)
);

CREATE TABLE TEAM_PS (
  TEAM_ID_PS                INT           NOT NULL,
  CLUB_ID_PS                INT           NOT NULL,
```

SQL File 3* | club_ps | division_ps | registration_ps | season_ps | team_ps | SQL File 8*

Limit to 1000 rows

```sql
  DIVISION_ID_PS          INT          NOT NULL,
  BIRTH_YEAR_PS           SMALLINT,
  PLAYER_SEX_PS           ENUM('M','F'),
  TEAM_NAME_PS            VARCHAR(20),
  TEAM_CONTACT_EMAIL_PS   VARCHAR(50),
  TEAM_CONTACT_PHONE_PS   CHAR(11),
  PRIMARY KEY (TEAM_ID_PS),
  CONSTRAINT FK_TEAM_PS_CLUB_PS
    FOREIGN KEY (CLUB_ID_PS)
    REFERENCES CLUB_PS(CLUB_ID_PS),
  CONSTRAINT FK_TEAM_PS_DIVISION_PS
    FOREIGN KEY (DIVISION_ID_PS)
    REFERENCES DIVISION_PS(DIVISION_ID_PS)
);

CREATE TABLE REGISTRATION_PS (
  REGISTRATION_ID_PS      INT          NOT NULL,
  TEAM_ID_PS              INT          NOT NULL,
  DIVISION_ID_PS          INT          NOT NULL,
  REG_DATE_PS             DATE,
```

SQL File 3* | club_ps | division_ps | registration_ps | season_ps | team_ps | SQL File 8*

Limit to 1000 rows

```sql
    FOREIGN KEY (TEAM_ID_PS)
    REFERENCES TEAM_PS(TEAM_ID_PS),
  CONSTRAINT FK_REG_PS_DIVISION_PS
    FOREIGN KEY (DIVISION_ID_PS)
    REFERENCES DIVISION_PS(DIVISION_ID_PS)
);

-- S5 table
CREATE TABLE GAME_PS (
  GAME_ID_PS              INT          NOT NULL,
  DIVISION_ID_PS          INT          NOT NULL,
  GAME_DATE_PS            DATE         NOT NULL,
  GAME_TIME_PS            TIME         NOT NULL,
  GAME_ZIPCODE_PS         CHAR(5)      NOT NULL,
  PRIMARY KEY (GAME_ID_PS),
  CONSTRAINT FK_GAME_PS_DIVISION_PS
    FOREIGN KEY (DIVISION_ID_PS)
    REFERENCES DIVISION_PS(DIVISION_ID_PS)
);
SHOW TABLES;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Tables_in_soccer_league_ps
club_ps
division_ps
game_ps
registration_ps
season_ps
team_ps

Result 7 | Result 8 | Result 9 | Result 10

Output

Action Output

#	Time	Action	Message
6	11:00:28	CREATE TABLE SEASON_PS (SEASON_ID_PS INT AUTO_INCREMENT PRIMARY KEY, SEASON...	0 row(s) affected
7	11:00:28	CREATE TABLE DIVISION_PS (DIVISION_ID_PS INT AUTO_INCREMENT PRIMARY KEY, SEASON...	0 row(s) affected
8	11:00:28	CREATE TABLE CLUB_PS (CLUB_ID_PS INT AUTO_INCREMENT PRIMARY KEY, CLUB_NAME_...	0 row(s) affected
9	11:00:28	CREATE TABLE TEAM_PS (TEAM_ID_PS INT AUTO_INCREMENT PRIMARY KEY, CLUB_ID_PS ...	0 row(s) affected
10	11:00:28	CREATE TABLE REGISTRATION_PS (REGISTRATION_ID_PS INT AUTO_INCREMENT PRIMARY KEY...	0 row(s) affected
11	11:00:28	SHOW TABLES	5 row(s) returned
12	11:00:28	DESCRIBE SEASON_PS	7 row(s) returned
13	11:00:28	DESCRIBE DIVISION_PS	7 row(s) returned
14	11:00:28	DESCRIBE CLUB_PS	5 row(s) returned
15	11:00:28	DESCRIBE TEAM_PS	7 row(s) returned
16	11:00:28	DESCRIBE REGISTRATION_PS	4 row(s) returned

TASK 4 – ADD DATA

SQL File 3* | club_ps | division_ps | registration_ps | season_ps | team_ps | SQL File 8*

Limit to 1000 rows

```sql
-- Task 4 - Add data by Priant Singh

INSERT INTO CLUB_PS
  (CLUB_ID_PS, CLUB_NAME_PS, CLUB_CONTACT_EMAIL_PS, CLUB_CONTACT_PHONE_PS, ABBREVIATION_PS)
VALUES
  (1, 'North Stars', 'north@club.com', '5710000001', 'NS'),
  (2, 'South Stars', 'south@club.com', '5710000002', 'SS');

INSERT INTO SEASON_PS
  (SEASON_ID_PS, SEASON_NAME_PS, YEAR_PS,
   REG_START_DATE_PS, REG_END_DATE_PS,
   SEASON_START_DATE_PS, SEASON_END_DATE_PS)
VALUES
  (1, 'Summer', 2025,
   '2025-04-01', '2025-05-15',
   '2025-06-01', '2025-08-01');

INSERT INTO DIVISION_PS
  (DIVISION_ID_PS, SEASON_ID_PS, AGE_GROUP_PS, PLAYER_SEX_PS, RANK_PS,
   DIV_CONTACT_EMAIL_PS, DIVISION_CONTACT_PHONE_PS)
```

SQL File 3* club_ps division_ps registration_ps season_ps team_ps SQL File 8*

Limit to 1000 rows

```
VALUES
  (10, 1, 'U12', 'M', 1,  'u12m@league.com', '5711111111'),
  (20, 1, 'U14', 'F', 1,  'u14f@league.com', '5712222222');

INSERT INTO TEAM_PS
  (TEAM_ID_PS, CLUB_ID_PS, DIVISION_ID_PS,
   BIRTH_YEAR_PS, PLAYER_SEX_PS, TEAM_NAME_PS,
   TEAM_CONTACT_EMAIL_PS, TEAM_CONTACT_PHONE_PS)
VALUES
  (101, 1, 10, 2013, 'M', 'Lions',  'lions@club.com', '7031111111'),
  (102, 1, 10, 2013, 'M', 'Tigers', 'tigers@club.com','7032222222'),
  (103, 1, 10, 2013, 'M', 'Bears',  'bears@club.com','7033333333'),
  (201, 2, 20, 2011, 'F', 'Eagles','eagles@club.com','7034444444'),
  (202, 2, 20, 2011, 'F', 'Hawks', 'hawks@club.com','7035555555'),
  (203, 2, 20, 2011, 'F', 'Falcons','falcons@club.com','7036666666');

SELECT * FROM CLUB_PS;
SELECT * FROM SEASON_PS;
SELECT * FROM DIVISION_PS;
SELECT * FROM TEAM_PS;
```

SQL File 3* club_ps division_ps registration_ps season_ps team_ps SQL File 8*

Limit to 1000 rows

```
  (201, 2, 20, 2011, 'F', 'Eagles','eagles@club.com','7034444444'),
  (202, 2, 20, 2011, 'F', 'Hawks', 'hawks@club.com','7035555555'),
  (203, 2, 20, 2011, 'F', 'Falcons','falcons@club.com','7036666666');

SELECT * FROM CLUB_PS;
SELECT * FROM SEASON_PS;
SELECT * FROM DIVISION_PS;
SELECT * FROM TEAM_PS;

INSERT INTO GAME_PS
  (GAME_ID_PS, DIVISION_ID_PS, GAME_DATE_PS, GAME_TIME_PS, GAME_ZIPCODE_PS)
VALUES
  (90000001, 10, '2025-06-01', '09:00:00', '22030'),
  (90000002, 10, '2025-06-02', '11:00:00', '22030'),
  (90000003, 10, '2025-06-03', '13:00:00', '22030'),
  (90000004, 20, '2025-06-01', '10:00:00', '22015'),
  (90000005, 20, '2025-06-02', '12:00:00', '22015');

SELECT * FROM GAME_PS ORDER BY GAME_ID_PS;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

CLUB_ID_PS	CLUB_NAME_PS	CLUB_CONTACT_EMAIL_PS	CLUB_CONTACT_PHONE_PS	ABBREVIATION_PS
1	North Stars	north@club.com	5710000001	NS
2	South Stars	south@club.com	5710000002	SS
NULL	NULL	NULL	NULL	NULL

Output

Action Output

#	Time	Action	Message
192	11:52:09	CREATE TABLE GAME_PS (GAME_ID_PS INT NOT NULL, DIVISION_ID_PS INT ...	0 row(s) affected
193	11:52:09	SHOW TABLES	6 row(s) returned
194	11:52:09	DESCRIBE GAME_PS	5 row(s) returned
195	11:52:09	INSERT INTO CLUB_PS (CLUB_ID_PS, CLUB_NAME_PS, CLUB_CONTACT_EMAIL_PS, CLUB_CONTACT...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
196	11:52:09	INSERT INTO SEASON_PS (SEASON_ID_PS, SEASON_NAME_PS, YEAR_PS, REG_START_DATE_PS,...	1 row(s) affected
197	11:52:09	INSERT INTO DIVISION_PS (DIVISION_ID_PS, SEASON_ID_PS, AGE_GROUP_PS, PLAYER_SEX_PS, RA...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
198	11:52:09	INSERT INTO TEAM_PS (TEAM_ID_PS, CLUB_ID_PS, DIVISION_ID_PS, BIRTH_YEAR_PS, PLAYER_S...	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0
199	11:52:09	SELECT * FROM CLUB_PS LIMIT 0, 1000	2 row(s) returned
200	11:52:09	SELECT * FROM SEASON_PS LIMIT 0, 1000	1 row(s) returned
201	11:52:09	SELECT * FROM DIVISION_PS LIMIT 0, 1000	2 row(s) returned
202	11:52:09	SELECT * FROM TEAM_PS LIMIT 0, 1000	6 row(s) returned
203	11:52:09	INSERT INTO GAME_PS (GAME_ID_PS, DIVISION_ID_PS, GAME_DATE_PS, GAME_TIME_PS, GAME_ZI...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
204	11:52:09	SELECT * FROM GAME_PS ORDER BY GAME_ID_PS LIMIT 0, 1000	5 row(s) returned

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

SEASON_ID_PS	SEASON_NAME_PS	YEAR_PS	REG_START_DATE_PS	REG_END_DATE_PS	SEASON_START_DATE_PS	SEASON_END_DATE_PS
1	Summer	2025	2025-04-01	2025-05-15	2025-06-01	2025-08-01
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

DIVISION_ID_PS	SEASON_ID_PS	AGE_GROUP_PS	PLAYER_SEX_PS	RANK_PS	DIV_CONTACT_EMAIL_PS	DIVISION_CONTACT_PHONE_PS
10	1	U12	M	1	u12m@league.com	5711111111
20	1	U14	F	1	u14f@league.com	5712222222
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

TEAM_ID_PS	CLUB_ID_PS	DIVISION_ID_PS	BIRTH_YEAR_PS	PLAYER_SEX_PS	TEAM_NAME_PS	TEAM_CONTACT_EMAIL_PS	TEAM_CONTACT_PHONE_PS
101	1	10	2013	M	Lions	lions@club.com	7031111111
102	1	10	2013	M	Tigers	tigers@club.com	7032222222
103	1	10	2013	M	Bears	bears@club.com	7033333333
201	2	20	2011	F	Eagles	eagles@club.com	7034444444
202	2	20	2011	F	Hawks	hawks@club.com	7035555555
203	2	20	2011	F	Falcons	falcons@club.com	7036666666
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

GAME_ID_PS	DIVISION_ID_PS	GAME_DATE_PS	GAME_TIME_PS	GAME_ZIPCODE_PS
90000001	10	2025-06-01	09:00:00	22030
90000002	10	2025-06-02	11:00:00	22030
90000003	10	2025-06-03	13:00:00	22030
90000004	20	2025-06-01	10:00:00	22015
90000005	20	2025-06-02	12:00:00	22015
NULL	NULL	NULL	NULL	NULL

Result 45 | Result 46 | Result 47 | Result 48 | CLUB_PS 49 | SEASON_PS 50 | DIVISION_PS 51 | TEAM_PS 52 | GAME_PS 53

Output

Action Output

#	Time	Action	Message
195	11:52:09	INSERT INTO CLUB_PS (CLUB_ID_PS, CLUB_NAME_PS, CLUB_CONTACT_EMAIL_PS, CLUB_CONTACT...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
196	11:52:09	INSERT INTO SEASON_PS (SEASON_ID_PS, SEASON_NAME_PS, YEAR_PS, REG_START_DATE_PS,...	1 row(s) affected
197	11:52:09	INSERT INTO DIVISION_PS (DIVISION_ID_PS, SEASON_ID_PS, AGE_GROUP_PS, PLAYER_SEX_PS, RA...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
198	11:52:09	INSERT INTO TEAM_PS (TEAM_ID_PS, CLUB_ID_PS, DIVISION_ID_PS, BIRTH_YEAR_PS, PLAYER_S...	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0
199	11:52:09	SELECT * FROM CLUB_PS LIMIT 0, 1000	2 row(s) returned
200	11:52:09	SELECT * FROM SEASON_PS LIMIT 0, 1000	1 row(s) returned
201	11:52:09	SELECT * FROM DIVISION_PS LIMIT 0, 1000	2 row(s) returned
202	11:52:09	SELECT * FROM TEAM_PS LIMIT 0, 1000	6 row(s) returned
203	11:52:09	INSERT INTO GAME_PS (GAME_ID_PS, DIVISION_ID_PS, GAME_DATE_PS, GAME_TIME_PS, GAME_ZI...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
204	11:52:09	SELECT * FROM GAME_PS ORDER BY GAME_ID_PS LIMIT 0, 1000	5 row(s) returned

TASK 5 – QUERY THE DATA

SQL File 3* | club_ps | division_ps | registration_ps | season_ps | team_ps | SQL File 8*

Limit to 1000 rows

```sql
-- Task 5 - Query the data by Priant Singh

-- 5.1 Games per ZIP code
SELECT
  GAME_ZIPCODE_PS,
  COUNT(*) AS games_count
FROM GAME_PS
GROUP BY GAME_ZIPCODE_PS
ORDER BY GAME_ZIPCODE_PS;

-- 5.2 Players per club
SELECT
  c.CLUB_NAME_PS,
  COUNT(p.PLAYER_ID_PS) AS total_players
FROM CLUB_PS c
JOIN TEAM_PS   t ON t.CLUB_ID_PS   = c.CLUB_ID_PS
JOIN PLAYER_PS p ON p.TEAM_ID_PS   = t.TEAM_ID_PS
GROUP BY c.CLUB_NAME_PS
ORDER BY c.CLUB_NAME_PS;
```

SQL File 3* | club_ps | division_ps | registration_ps | season_ps | team_ps | SQL File 8*

Limit to 1000 rows

```sql
ORDER BY c.CLUB_NAME_PS;

-- 5.3 Players per division (current season)
SELECT
  d.AGE_GROUP_PS AS division,
  COUNT(p.PLAYER_ID_PS) AS total_players
FROM DIVISION_PS d
JOIN TEAM_PS   t ON t.DIVISION_ID_PS = d.DIVISION_ID_PS
JOIN PLAYER_PS p ON p.TEAM_ID_PS     = t.TEAM_ID_PS
WHERE d.SEASON_ID_PS = 1   -- Summer 2025
GROUP BY d.AGE_GROUP_PS;

-- 5.4 Teams per coach
SELECT
  CONCAT(co.FIRST_NAME_PS, ' ', co.LAST_NAME_PS) AS coach_name,
  COUNT(t.TEAM_ID_PS) AS teams_count
FROM COACH_PS co
JOIN TEAM_PS  t  ON t.COACH_ID_PS = co.COACH_ID_PS
GROUP BY coach_name
ORDER BY coach_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

GAME_ZIPCODE_PS	games_count
22015	2
22030	3

ult 69 | CLUB_PS 70 | SEASON_PS 71 | DIVISION_PS 72 | TEAM_PS 73 | COACH_PS 74 | PLAYER_PS 75 | GAME_PS 76 | Result 77 | Result 78 | Result 79 | Result 8 | Re

Output

Action Output

#	Time	Action	Message
253	12:03:04	INSERT INTO GAME_PS (GAME_ID_PS, DIVISION_ID_PS, GAME_DATE_PS, GAME_TIME_PS, GAME_ZI...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
254	12:03:04	SELECT * FROM CLUB_PS LIMIT 0, 1000	2 row(s) returned
255	12:03:04	SELECT * FROM SEASON_PS LIMIT 0, 1000	1 row(s) returned
256	12:03:04	SELECT * FROM DIVISION_PS LIMIT 0, 1000	2 row(s) returned
257	12:03:04	SELECT * FROM TEAM_PS LIMIT 0, 1000	6 row(s) returned
258	12:03:04	SELECT * FROM COACH_PS LIMIT 0, 1000	2 row(s) returned
259	12:03:04	SELECT * FROM PLAYER_PS LIMIT 0, 1000	11 row(s) returned
260	12:03:04	SELECT * FROM GAME_PS ORDER BY GAME_ID_PS LIMIT 0, 1000	5 row(s) returned
261	12:03:04	SELECT GAME_ZIPCODE_PS, COUNT(*) AS games_count FROM GAME_PS GROUP BY GAME_ZIPCOD...	2 row(s) returned
262	12:03:04	SELECT c.CLUB_NAME_PS, COUNT(p.PLAYER_ID_PS) AS total_players FROM CLUB_PS c JOIN TEAM_...	2 row(s) returned
263	12:03:04	SELECT d.AGE_GROUP_PS AS division, COUNT(p.PLAYER_ID_PS) AS total_players FROM DIVISION_PS ...	2 row(s) returned
264	12:03:04	SELECT CONCAT(co.FIRST_NAME_PS, ' ', co.LAST_NAME_PS) AS coach_name, COUNT(t.TEAM_ID_PS...	2 row(s) returned

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CLUB_NAME_PS	total_players
North Stars	6
South Stars	5

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

division	total_players
U12	6
U14	5

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

coach_name	teams_count
Alice Smith	3
Bob Jones	3

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

COACH_ID_PS	FIRST_NAME_PS	LAST_NAME_PS
301	Alice	Smith
302	Bob	Jones
NULL	NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

PLAYER_ID_PS	FIRST_NAME_PS	LAST_NAME_PS	TEAM_ID_PS
401	Carlos	Diaz	101
402	Dana	Wu	101
403	Ethan	Lee	101
404	Fiona	Ng	102
405	Gabe	O'Neil	102
406	Hana	Patel	103
407	Ivan	Qin	201
408	Jade	Reyes	201
409	Kira	Shah	202
410	Liam	Torres	202
411	Maya	Upton	203
NULL	NULL	NULL	NULL